



United International University

Department of Computer Science and Engineering

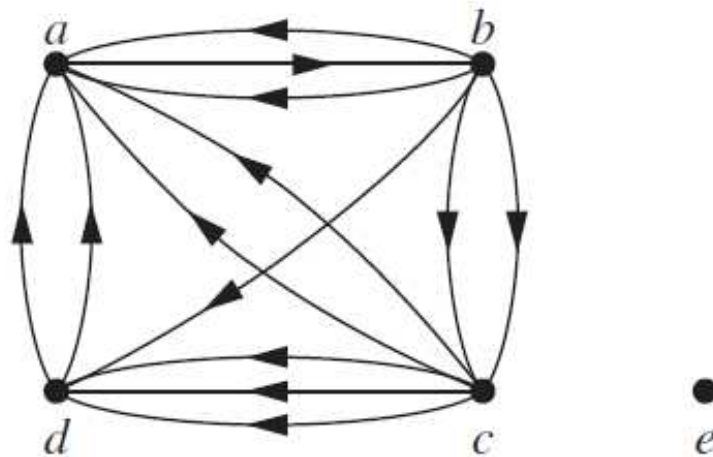
CSE 2213/CSI 219: Discrete Mathematics (BSCSE and BSDS)

Final-term Examination Fall 2025

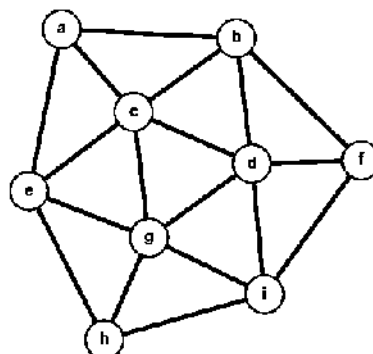
Total Marks: 40 Time: 2 hour

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

1. (a) A university wants to form a committee from 4 teachers and 4 students. [2+2]
 - i. In how many ways can a committee of 3 members be selected if the committee must contain at least one teacher?.
 - ii. In how many different ways can the 3 selected members be arranged in a line for a group photograph if exactly one of them is a teacher?
- (b) A student is planning a university tour by bus. They must choose one of four bus companies, a seat type (AC or Non-AC), and a meal. There are three meal options, but only two are available with an AC seat. How many different travel plans are possible? [4]
- (c) A library has 10 shelves, and books can be placed among the shelves in any way. What is the minimum number of books that must be placed in the library to guarantee that at least one shelf contains 15 or more books? [4]
2. (a) During a chess tournament with 8 players, each player plays against every other player exactly once. After the tournament, the organizer wants to draw a diagram showing who beat whom. What kind of graph do we get? Explain. [2]
- (b) Consider a simple graph with 7 vertices where every vertex has degree 6. What kind of graph is this? Give its standard name and explain briefly why. [2]
- (c) Does the following graph follow the Handshaking Theorem? Justify your answer with brief explanation. [4]



3. (a) Determine whether the graph in the following figure is bipartite or not, using the 2-coloring algorithm. If the graph is bipartite, redraw the graph in bipartite form. [4]



(b) Consider the following adjacency matrix for a directed graph with 6 vertices. Draw the graph from the adjacency matrix. Find the in-degree and out-degree of each vertex. [4]

	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>
<i>A</i>	1	1	0	0	0	0
<i>B</i>	1	0	0	0	0	0
<i>C</i>	0	0	1	1	0	0
<i>D</i>	0	1	0	0	1	0
<i>E</i>	0	0	0	0	0	1
<i>F</i>	1	0	0	0	0	1

4. (a) Consider the following numbers: **32, 21, 24, 31, 10, 27, 35, 40, 38**
- Construct a Binary Search Tree (BST) using the above numbers. [3]
 - Determine the in-order and post-order traversal of the constructed BST. [3]
 - Identify the ancestors of 31 and the descendants of 35 from the constructed BST. [3]
- (b) A full 6-ary tree has a total of 43 vertices. How many leaves does the tree have? [3]